A FREE COMPLEX MAPPING TO A MODULE MAPS INTO ANY RESOLUTION, AND ANY TWO SUCH MAPS ARE HOMOTOPIC

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Allowed standard prerequisites (editorial).

- Lifting along a surjection out of a free module. Used repeatedly in the counted proof (lines 17740-17749). Produces the comparison maps degree by degree from freeness of the complex terms.
- Exactness of a resolution for lifting into kernels. Used in both parts of the counted proof (lines 17746 and 17759). Guarantees that whatever maps to zero can be lifted one degree further.

Lemma 0.1. *Let R be a ring. Let $M \to N$ be a map of R-modules. Let $N_\bullet \to N$ be an arbitrary resolution. Let*

$$\ldots \to F_2 \to F_1 \to F_0 \to M$$

be a complex of R-modules where each F_i is a free R-module. Then

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(1) *there exists a map of complexes $F_\bullet \to N_\bullet$ such that*

$$\begin{array}{ccc} F_0 & \longrightarrow & M \\ \downarrow & & \downarrow \\ N_0 & \longrightarrow & N \end{array}$$

is commutative, and

(2) *any two maps $\alpha, \beta : F_\bullet \to N_\bullet$ as in (1) are homotopic.*

Proof. Proof of (1). Because F_0 is free we can find a map $F_0 \to N_0$ lifting the map $F_0 \to M \to N$. We obtain an induced map $F_1 \to F_0 \to N_0$ which ends up in the image of $N_1 \to N_0$. Since F_1 is free we may lift this to a map $F_1 \to N_1$. This in turn induces a map $F_2 \to F_1 \to N_1$ which maps to zero into N_0. Since $N_\bullet$ is exact we see that the image of this map is contained in the image of $N_2 \to N_1$. Hence we may lift to get a map $F_2 \to N_2$. Repeat.

Proof of (2). To show that α, β are homotopic it suffices to show the difference $\gamma = \alpha - \beta$ is homotopic to zero. Note that the image of $\gamma_0 : F_0 \to N_0$ is contained in the image of $N_1 \to N_0$. Hence we may lift γ_0 to a map $h_0 : F_0 \to N_1$. Consider the map $\gamma_1' = \gamma_1 - h_0 \circ d_{F,1}$. By our choice of h_0 we see that the image of γ_1' is contained in the kernel of $N_1 \to N_0$. Since $N_\bullet$ is exact we may lift γ_1' to a map $h_1 : F_1 \to N_2$. At this point we have $\gamma_1 = h_0 \circ d_{F,1} + d_{N,2} \circ h_1$. Repeat. $\square$